

■ LangGraph Multi-Agent Pipeline

Job Application Assistant

A LangGraph-based multi-agent pipeline for tailored job applications

LangGraph

LangChain

OpenAI GPT-4o-mini

ChromaDB

Pydantic v2

Streamlit

Recruiters discard generic applications within 6 seconds

- Tailoring a resume + cover letter to a JD takes **1–3 hours** done properly
- Most candidates **skip it or do it inconsistently**
- It's actually a **pipeline of 4 dependent tasks**, not one task

1 Analyze JD vs resume — analytical

2 Rewrite CV for hidden relevance — creative + constrained

3 Write a grounded cover letter — generative + factual

4 Human review before anything is used — quality gate

KEY INSIGHT

No single LLM prompt can do all four well.

This is a **coordination problem** — it requires a multi-agent architecture, not just generation.

Goals & Target Users

GOALS & SUCCESS METRICS

Reduce tailoring time	~2 hrs manual → < 5 min
Keyword match rate	≥ 70% on strong-fit roles
Human control	0 outputs without approval
No hallucination	never fabricate experience
Transparency	logged + LangSmith traces

TARGET USERS

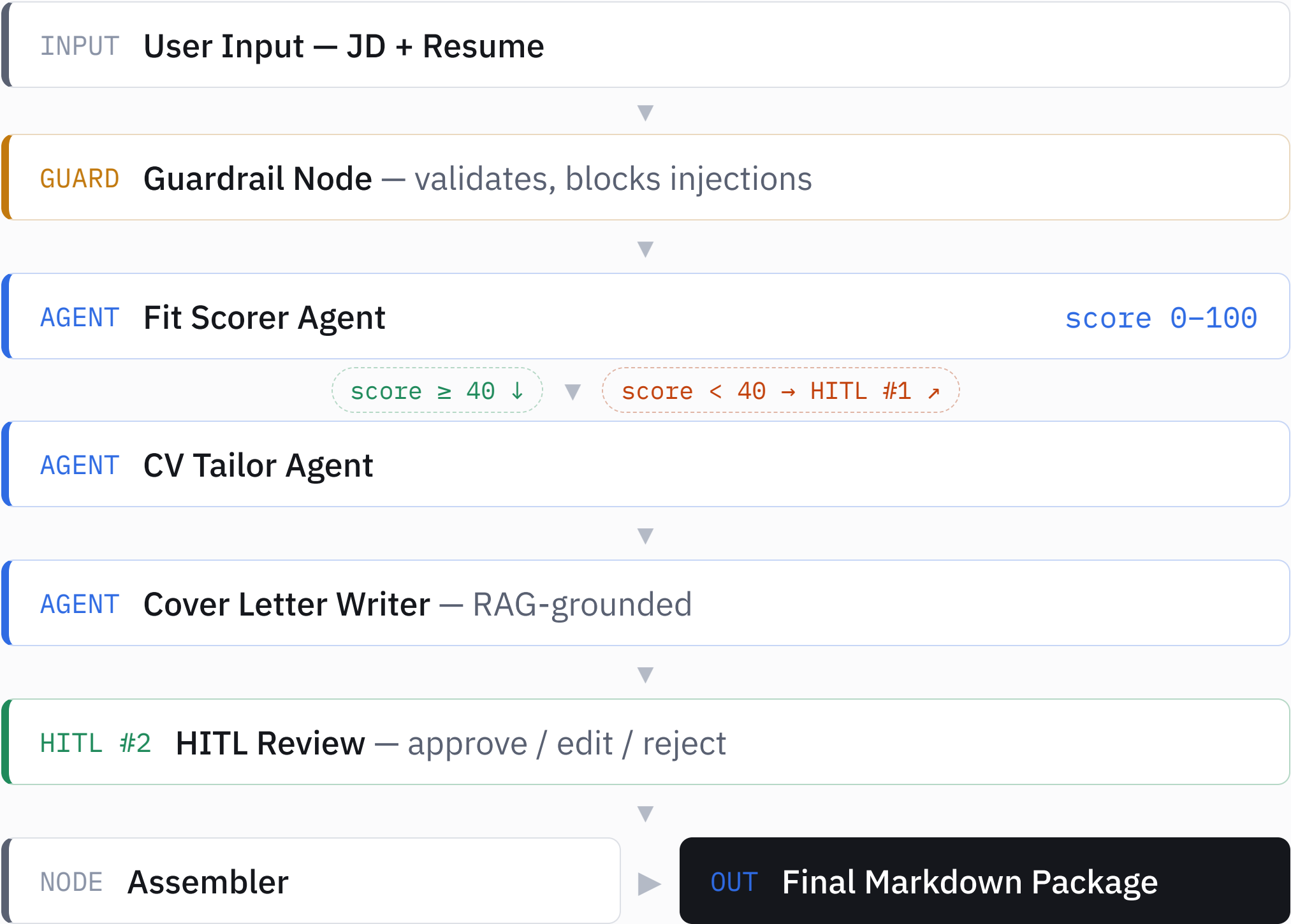
Job seekers — students, grads, career changers applying to competitive roles.

Secondary: career coaches reviewing AI drafts.

NON-GOALS

- Does not auto-submit applications
- Does not scrape job boards
- No persistent user data
- Not a general career chatbot

System Architecture



HITL #1 · CONDITIONAL

Low Fit Warning fires when **score < 40**. User can proceed anyway or abort.

HITL #2 · ALWAYS

Full review of every output before assembly — **nothing ships unreviewed.**

AT A GLANCE

- 7 graph nodes
- 4 conditional routing functions
- 2 human-in-the-loop interrupts

The Three Agents

	AGENT 01 Fit Scorer	AGENT 02 CV Tailor	AGENT 03 Cover Letter Writer
Model	gpt-4o-mini · temp 0.1	gpt-4o-mini · temp 0.3	gpt-4o-mini · temp 0.5
Tools	extract_keywords + RAG	rewrite_bullets	validate_cover_letter + RAG (2 queries)
Output	FitScoreOutput score, matched/missing keywords, strengths, gaps, recommendation	TailoredCVOutput summary, ≤6 tailored bullets, skills, changes made	CoverLetterOutput subject line, opening / body / closing paragraphs
Key constraint	Never crashes — falls back to score=50	Never fabricates experience	Self-corrects once if validation fails

Tools & RAG Knowledge Base

TOOLS deterministic · zero LLM cost · @tool

`extract_keywords(text)`
Rule-based skill / qualification extraction

`rewrite_bullets(bullets, keywords, role)`
Keyword-targeted rewrite suggestions

`validate_cover_letter(text)`
Checks word count (150–500), salutation, closing

RAG KNOWLEDGE BASE

7

embedded knowledge documents

Software engineering

Data science

Product management

Marketing

Cover letter best practices

Resume / ATS optimization

Startup vs enterprise strategy

Embedding model	text-embedding-3-small
Vector store	ChromaDB · persisted locally
Chunking	400 tokens / 50 overlap
Similarity search	<code>k = 3</code>

Two interrupt points — nothing ships without a human

HITL #1

CONDITIONAL · score < 40

Low Fit Warning

Shows the **score**, the **recommendation**, and the **gaps** the candidate would need to overcome.

USER CHOICES

Proceed anyway → CV Tailor

Abort → END

HITL #2

ALWAYS · after all 3 agents

Full Review

Shows fit score, keyword analysis, tailored CV and cover letter — **all editable inline.**

USER CHOICES

Approve

Edit + Approve

Reject → END

```
IMPLEMENTATION LangGraph interrupt_before=["hitl_review", "low_fit_warning"] · MemorySaver checkpointer
```

Guardrails & Safety

GUARD	IMPLEMENTATION	TRIGGER
Empty input	<code>strip()</code> check	blank JD or resume
Minimum length	word count	JD < 30 words or resume < 50 words
Prompt injection	string match · 8 known patterns	"ignore previous instructions", "jailbreak", ...
No fabrication	system prompt hard constraint	CV Tailor agent
HITL gates	graph interrupts	low fit + always before final output
No auto-send	architecture constraint	markdown download only, no external API calls

Evaluation Results

evals/run_evals.py · 5 cases

#	CASE	ASSERTION	COST
1	Happy path — strong match	score ≥ 60, all outputs populated, package assembled	API key
2	Empty resume	guardrail blocks before any agent runs	free
3	Prompt injection	detected, graph aborts, zero LLM calls	free
4	Low fit routing	score < 40, routes to low_fit_warning, graph pauses	API key
5	Structured output validation	all Pydantic models validate, ranges correct	API key

Evals **2 & 3 run free** (no LLM cost); evals **1, 4, 5 require** OPENAI_API_KEY.

Key Design Decisions

LangGraph > linear chain

Steps depend on prior structured output, routing is conditional on score, and two HITL pause/resume points require a **stateful graph**.

Why Pydantic

Malformed agent output raises `ValidationError` immediately instead of **silently corrupting state**.

Why rule-based tools

Deterministic, zero-cost, and they **reduce hallucination** vs. doing the work inside the LLM prompt.

TEAM MEMBERS

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TECH STACK

LangGraph ≥0.2.0

LangChain ≥0.3.0

OpenAI GPT-4o-mini

ChromaDB ≥0.5.0

Pydantic v2

Streamlit ≥1.38.0

Python 3.10+